California Polytechnic State University Pomona
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Circuit Analysis 2 Laboratory

ECE 2101L-03
Report #1

Introduction to Function Generator, Oscilloscope, & Matlab

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Introduction

In this lab, we will explore the ways in which we use function generators and oscilloscopes to create, analyze, measure, and visualize many aspects of a sinusoidal voltage waveform. Function generators as the name suggests can create various voltage functions that are useful in powering circuits using sine wave voltage and in digital logic as a simple clock cycle using square waves. Oscilloscopes on the other hand are used to visualize and measure these moving voltages that are not possible to analyze using traditional multimeters.

Objectives

- To practice basic operational controls of the function generator.
- To learn basic controls of the two-channel oscilloscope.

A. Pre-Lab Section

Task A1: Sinusoids

1. Given the following voltage

() = 15cos (377t + 19) Volt

a) What is the amplitude of v(t)?

   The amplitude is 15 volts.

b) What is the frequency measured in rad/s of v(t)?

   The frequency is 377 (rad/s).

c) What is the frequency measured in Hertz of v(t)?

   We get the frequency in Hertz by using 2πf = 377.

   From there we get f = 377/2π. F = 60Hz

d) What is the phase of v(t)?

   The phase is 19.

e) Calculate the period of v(t)

   Period is 1/f. So, T= 1/60. T = 0.0166 seconds

f) Calculate the RMS (i.e. the effective value) of v(t)

   Vrms = $\frac{Vm}{\sqrt{2}}$ = $\frac{15}{\sqrt{2}}$ = 10.607 volts

2. Figure 1 shows a graph of a sinusoidal voltage v(t). Clearly mark the following parameters in the figure

a) Amplitude (in yellow on fig 1)

b) Period (in green on fig 1)

c) Peak to peak value (in red on fig 1)

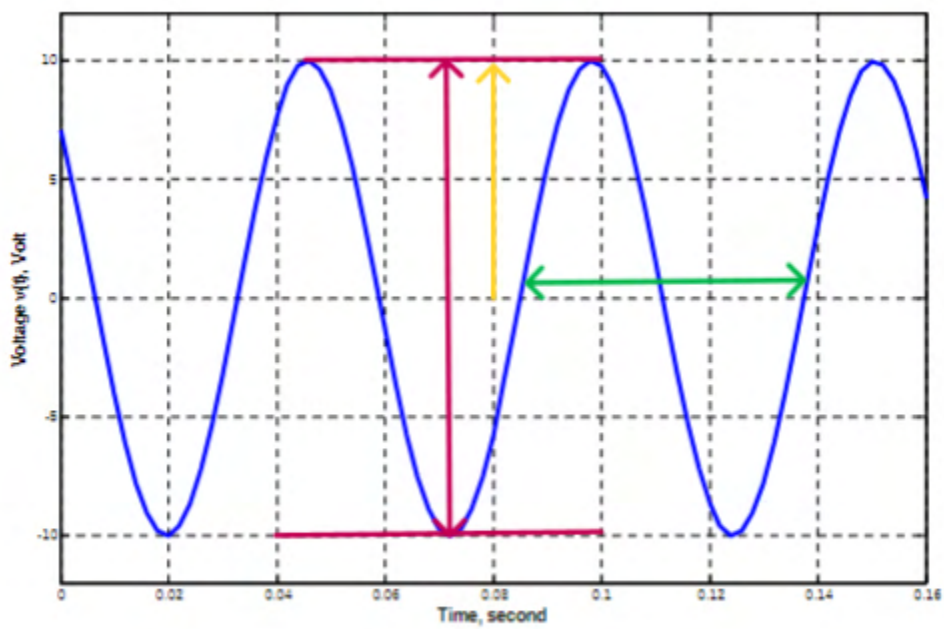

Fig. 1 Graph of sinusoidal voltage v(t)

Task A2: Matlab

Browse or search the Matlab Help content and answer the following questions.

1) What is a function in Matlab? Give two examples.

   A function in Matlab is a task where you put in input(s) and get a desired output. We can also create our own functions in order to reuse the same code multiple times thus reducing the length of a script and saving time while coding. One example is the **factorial**(n) function. When you use this function it will give you the factorial of your input. Another example of a function is the **nthroot**(x,n) function which will take the x input and root it by the n input.

2) Describe what the function **plot** does.

The function plot takes an X and a Y and plots them against each other in 2D. It can also graph data tables, sets of data points and a single data point.

3) Explain what an **M-file** is and how to create an M-file in Matlab.

   M-files are text files that save Matlab code and commands. M-files are useful when sending code to someone else since they can put it into Matlab and get the functioning code. To create an M-file you click on New (Labeled with a yellow plus near the top left) and a dropdown appears. Then click on the script option. This will open a text editor where you can enter Matlab functions. Once you are done click save, then save as, and lastly make sure the file type is “.m”

Task A3: Oscilloscope

Using the lab Agilent DSOX2022A Oscilloscope User's Guide, describe how to save the oscilloscope screen images to a USB storage device.

1. Press [Save/Recall] > Save > Format; then, turn the Entry knob to select 8-bit Bitmap image (*.bmp), 24-bit Bitmap image (*.bmp), or 24-bit image (*.png).
2. Press the softkey in the second position and use the Entry knob to navigate to the save location. See "To navigate storage locations" on page 234.
3. Press the Settings softkey. Here you can save other data points in a separate .txt file or, change the image properties such as its background and palette.
4. Finally, press the Press to Save softkey.

B. Lab Section:

Task B1: Working with the Function Generator

In this task we used the control panel of the function generator to specify a sinusoidal voltage and output it to the oscilloscope. We used our BNC to BNC cable and plugged one end into the OUTPUT of the function generator and the other end to Channel 1 of the oscilloscope.

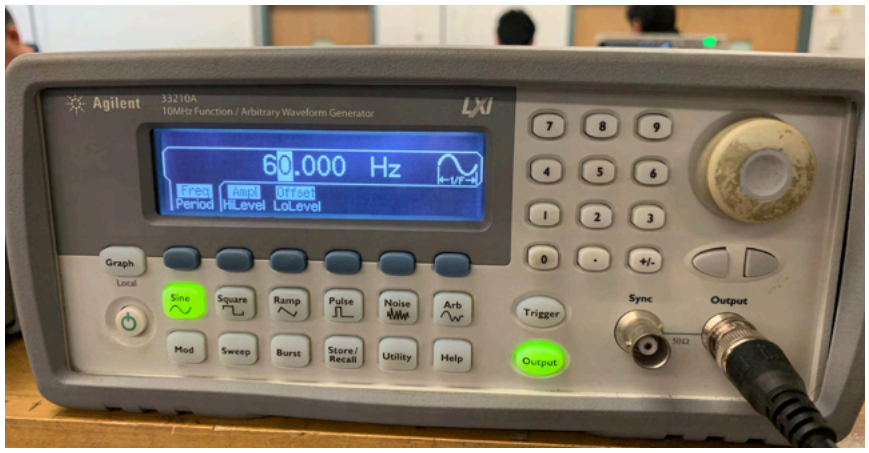

Figure 2A - Frequency of the sine function

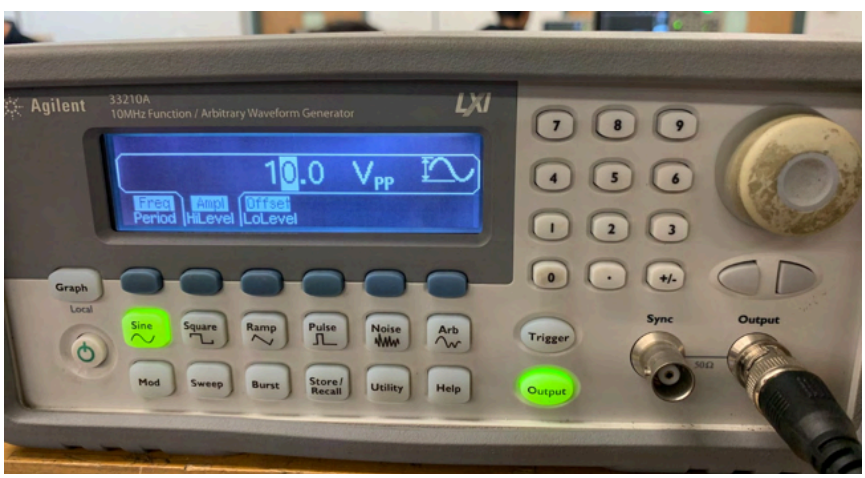

Figure 2B - Peak-to-peak voltage of the sine function

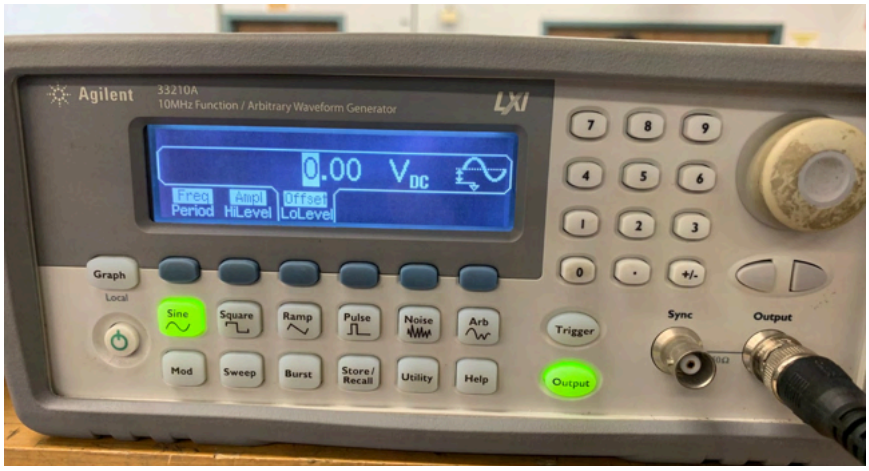

Figure 2C - DC offset voltage of the sine function

In Figure 2A, we selected the sine function and set the frequency to 60Hz. In Figure 2C, we set the DC offset to 0 volts and pressed the output. In Figure 2B, we set the peak-to-peak voltage to 10 volts. All three steps were done using the soft keys and the number pad on the right.

Task B2: Working with oscilloscope

In this task, we will present some of the basic tools and functions that we need to operate an oscilloscope in analyzing AC voltage curves. The figures below describe our step-by-step process of using some of the basic functions of the oscilloscope.

Task B2.1: Experiments involving Horizontal control system, Vertical control system, and Trigger system.

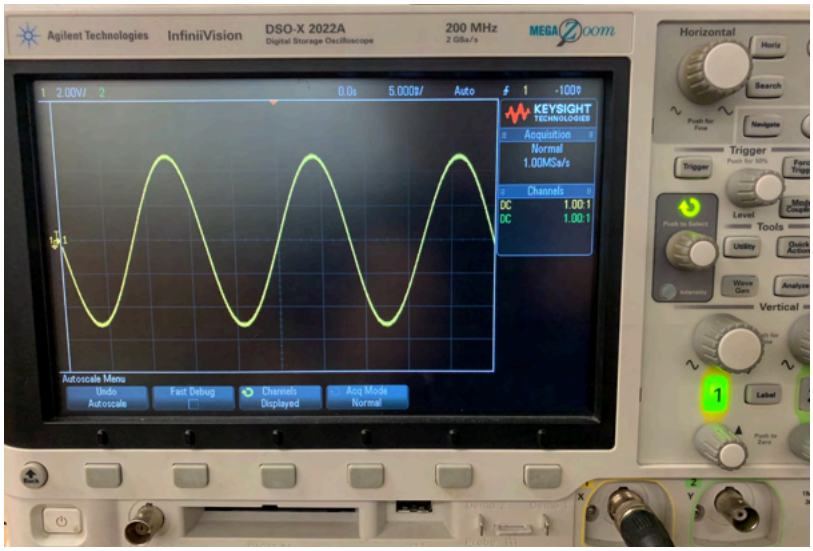

Figure 3 - A complete sine wave on the screen

In Figure 3, we made sure that our sine waveform was clear by utilizing the horizontal, and vertical control units and the auto-scale feature.

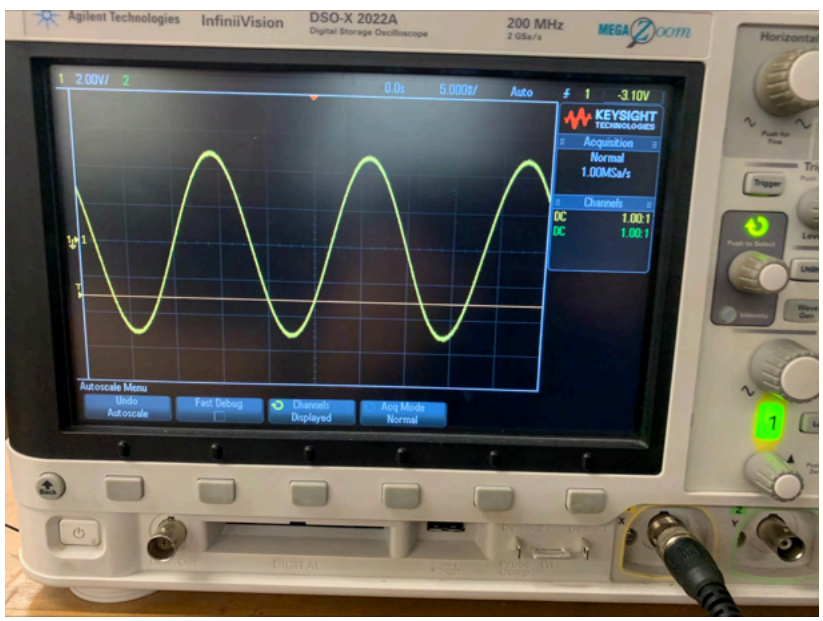

Figure 4A - Trigger level set at -3.1 volts

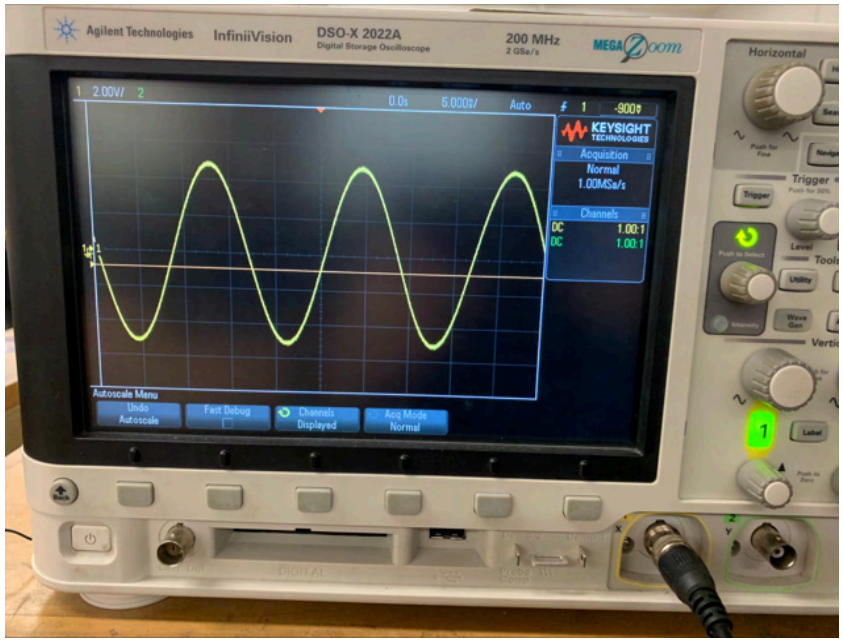

Figure 4B- Trigger level set at -0.9 volts

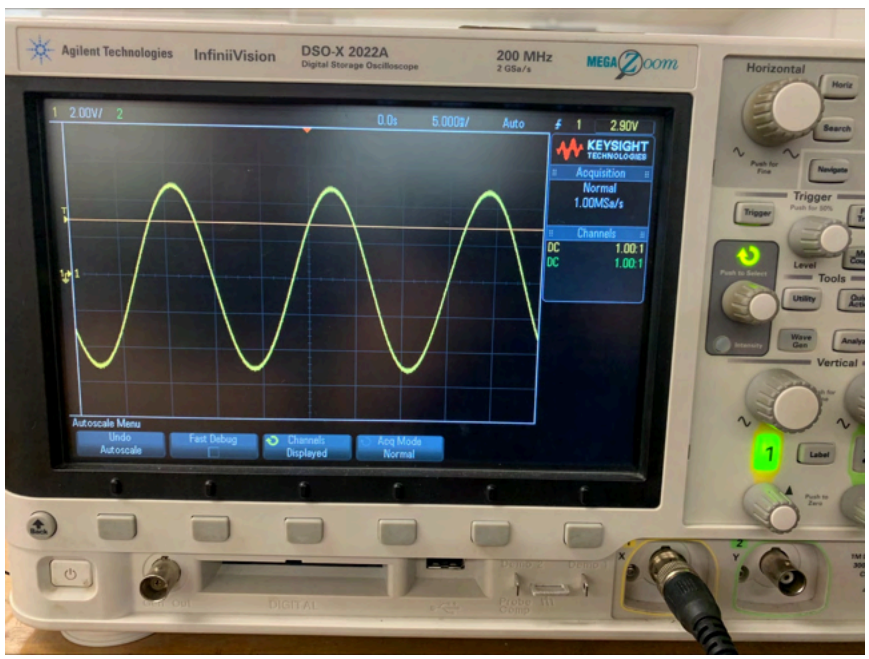

Figure 4C - Trigger level set at 2.9 volts

In Figure 4A, we utilized the Level knob in the Trigger control to see how the sinusoidal voltage changed. Here it is -3.1 volts. In Figure 4B, we utilized the Level knob to see how the sinusoidal voltage changed. Here it is -900 millivolts. In Figure 4C, we utilized the Level knob to see how

the sinusoidal voltage changed. Here it is 2.9 volts. In the above scenarios, we observe that the sine wave shifts left or right slightly as we move the trigger level up or down. The sine wave becomes unstable if the trigger level is set too low or too high thus becoming unreadable.

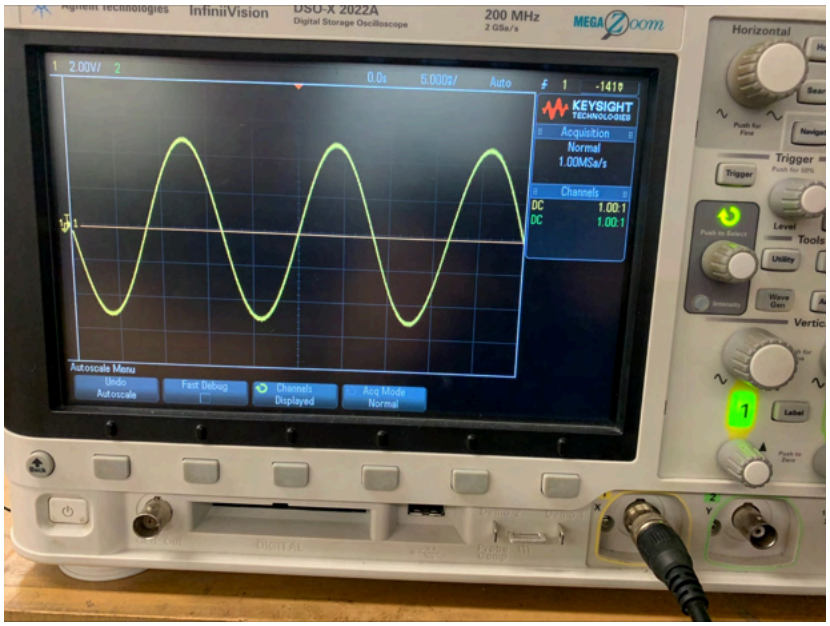

Figure 5 - Resetting the trigger level to 0 volts

In Figure 5, we pushed the Level knob once. We noticed that it reset the line to the middle of the sine waveform. I think it would be useful for resetting after checking different trigger levels or finding the exact middle which is 50% of the peak-to-peak voltage.

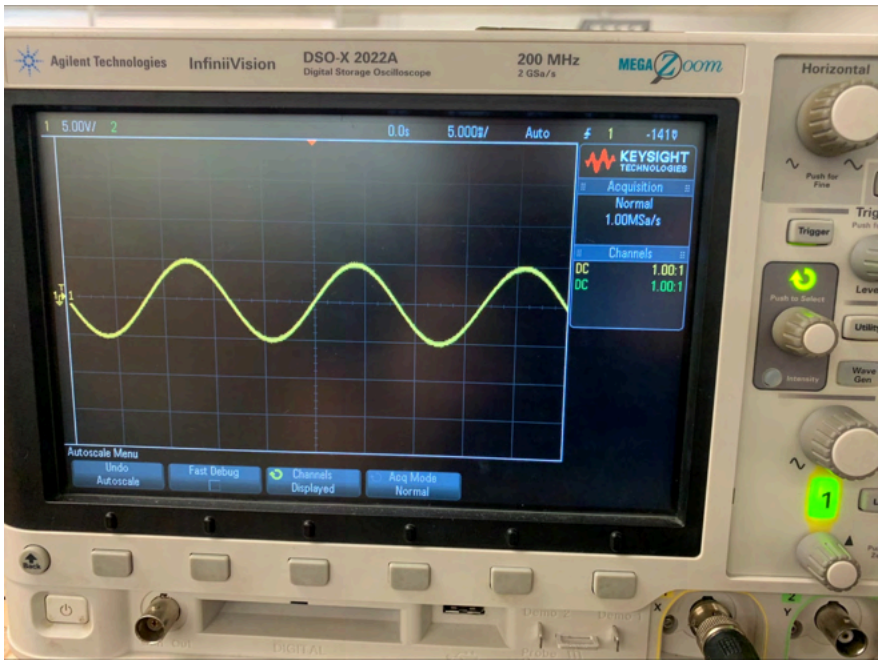

Figure 6A - Changing the vertical control

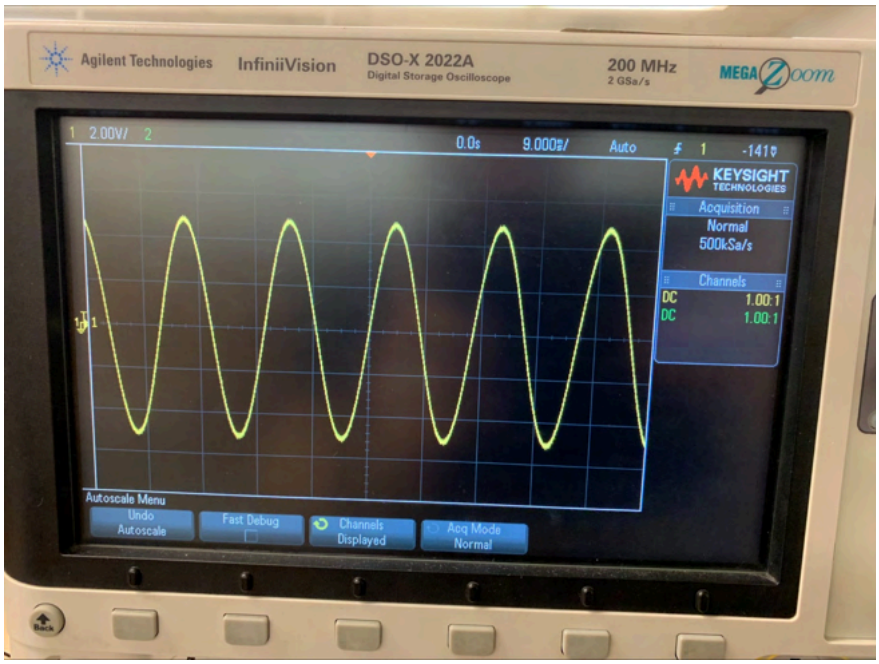

Figure 6B - Changing the horizontal control

In Figure 6A, we turned the Vertical control knob and saw the sine waveform change since it became shorter than it originally was. This is because the voltage per division got higher (from 2 volts/div to 5 volts/div) when we did this. In Figure 6B, we turned the Horizontal control knob

and saw the sine waveform get more compact. This is because the seconds per division got higher (from 5ms/div to 9ms/div) when we did this.

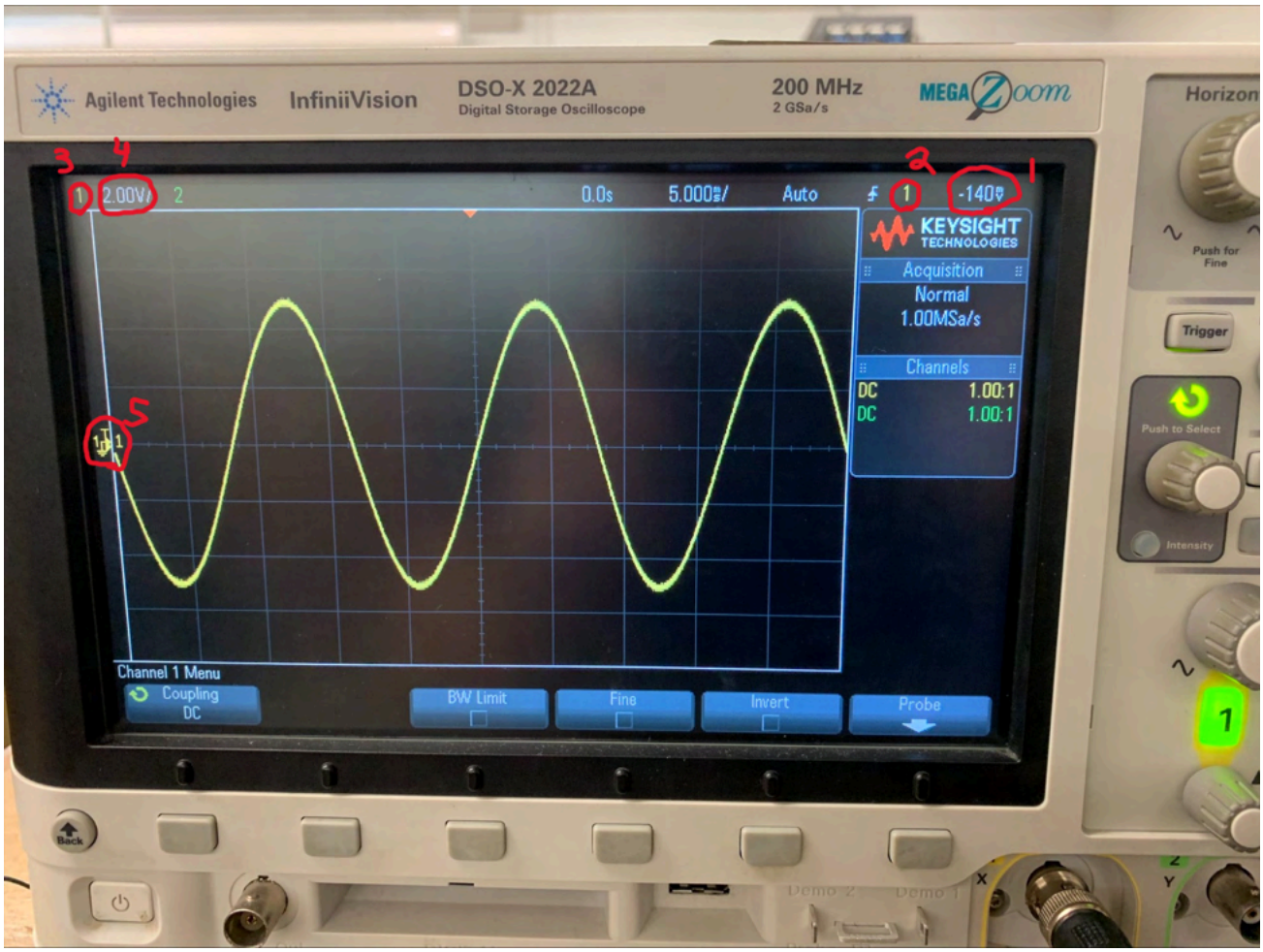

Figure 7 - The placement of display parameters

In Figure 7, we pressed the [1] channel key to bring up the Channel 1 menu. The trigger level (circle labeled 1) was -140 mV. The trigger source (circle labeled 2) was 1. The channel (circle labeled 3) was 1. The voltage/division (circle labeled 4) was 2 volts per division. Lastly, the channel/ground level is circled with the label 5.

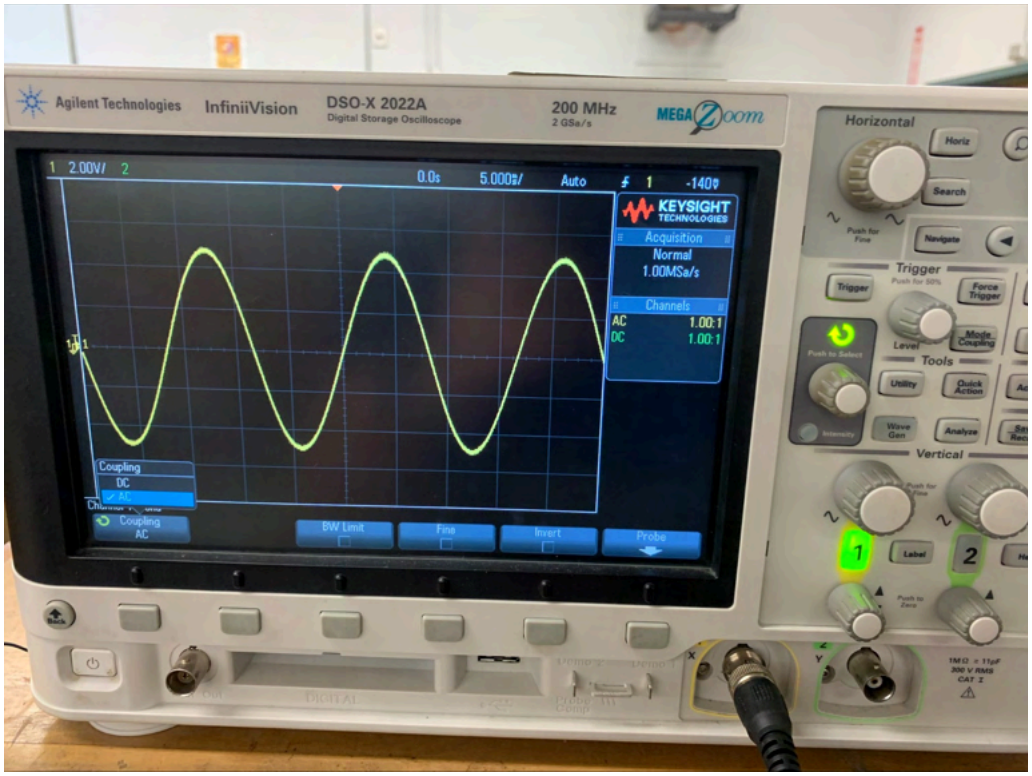

Figure 8 - Coupling menu

In Figure 8, we were able to switch the coupling from DC to AC. We did this by pressing the [1] channel key and then pressing the first softkey. Doing this allowed us to switch from DC to AC or vice versa.

Task B2.2: Experiment involving Measure system

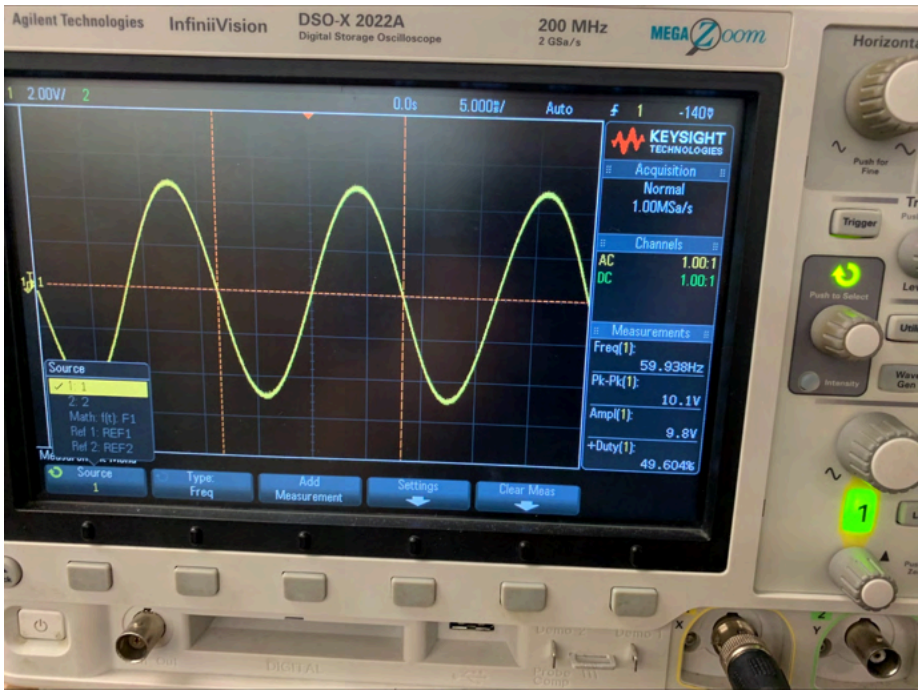

Figure 9A - Measurement lines and menu displayed on the screen

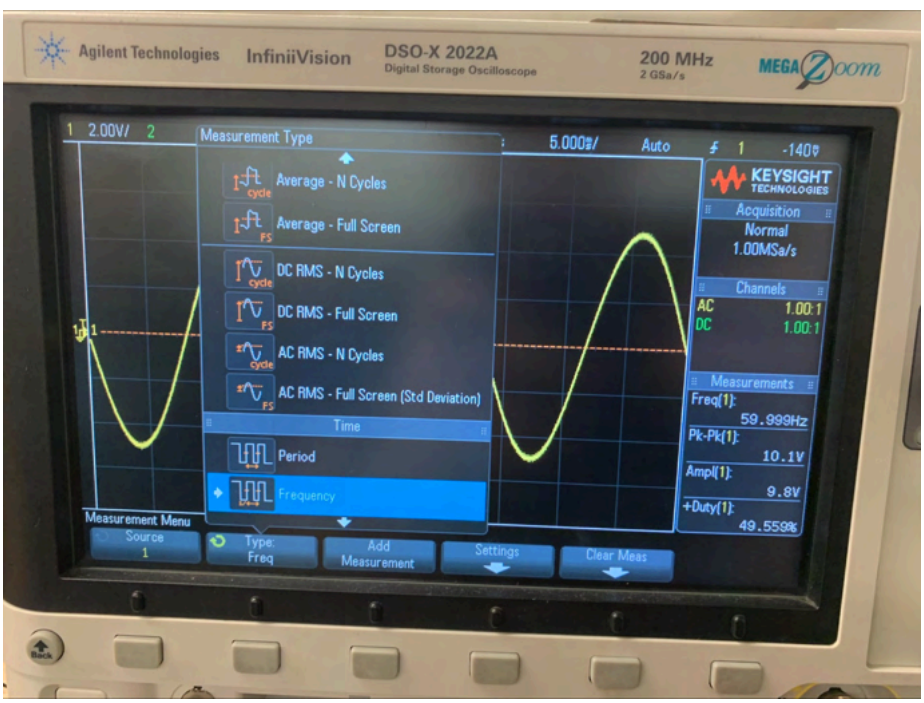

Figure 9B - Selecting the type of measurement to the frequency

In Figure 9A, we first pressed the [Meas] button and then selected Source 1. In Figure 9B, we

pressed the Type softkey, then selected Frequency, and lastly pressed the Add Measurement softkey. This displayed Freq(1) data point under Measurements on screen thus allowing us to take note of the sine wave's frequency.

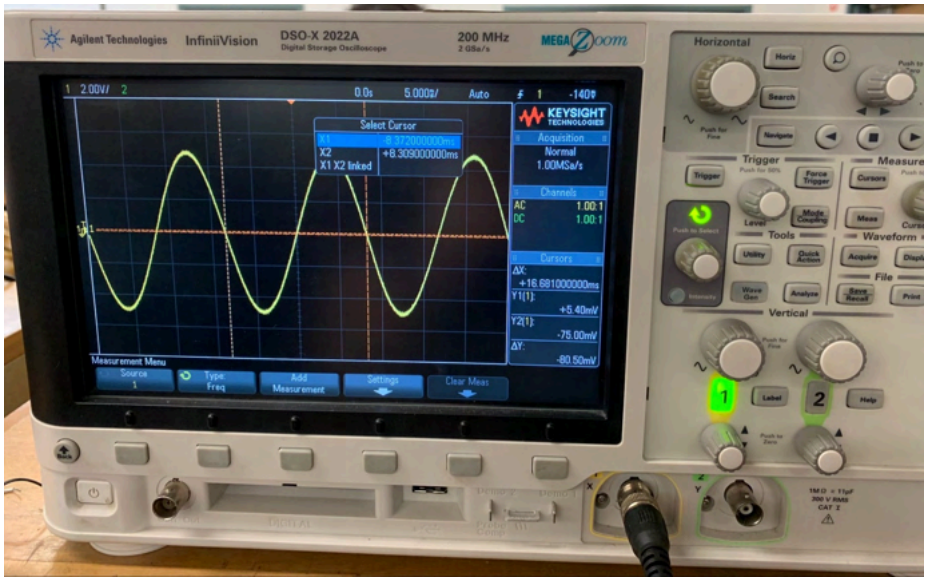

Figure 9C - Adjusting vertical measurement lines X1 and X2

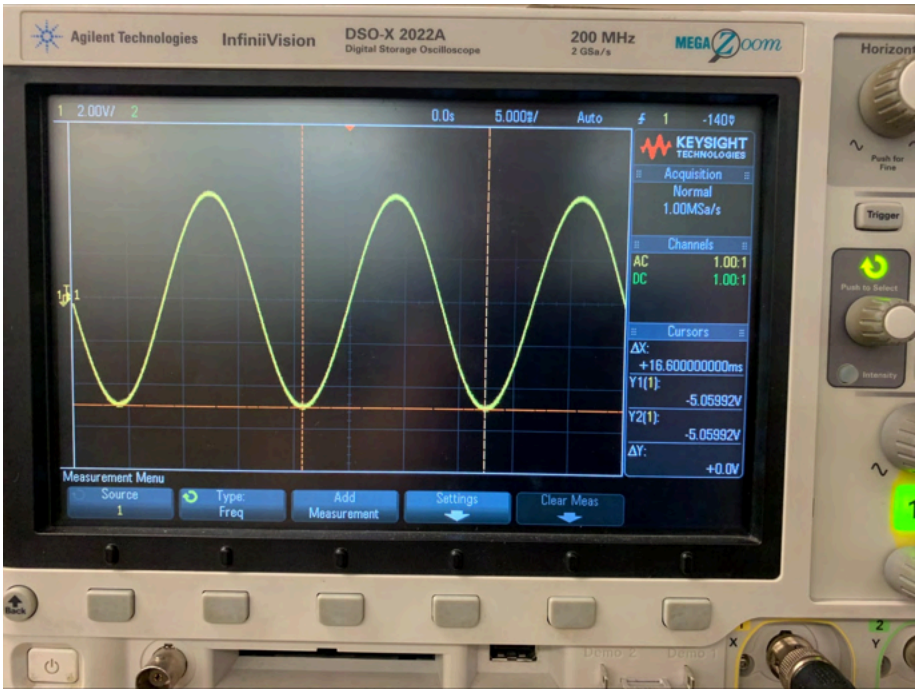

Figure 9D - Measurement lines X1 and X2 showing the period of the sine wave

In Figure 9C, we pushed the Cursors knob which opened a menu of cursor lines. Here, we can select X1, X2, or both in order to move them. In Figure 9D, we determined the period of sinusoidal voltage which was 16.6 ms. We did this by setting X1 and X2 to separate peaks next to each other and observing the ΔX parameter under Cursors.

Figure 10A - Selecting the AC RMS measurement

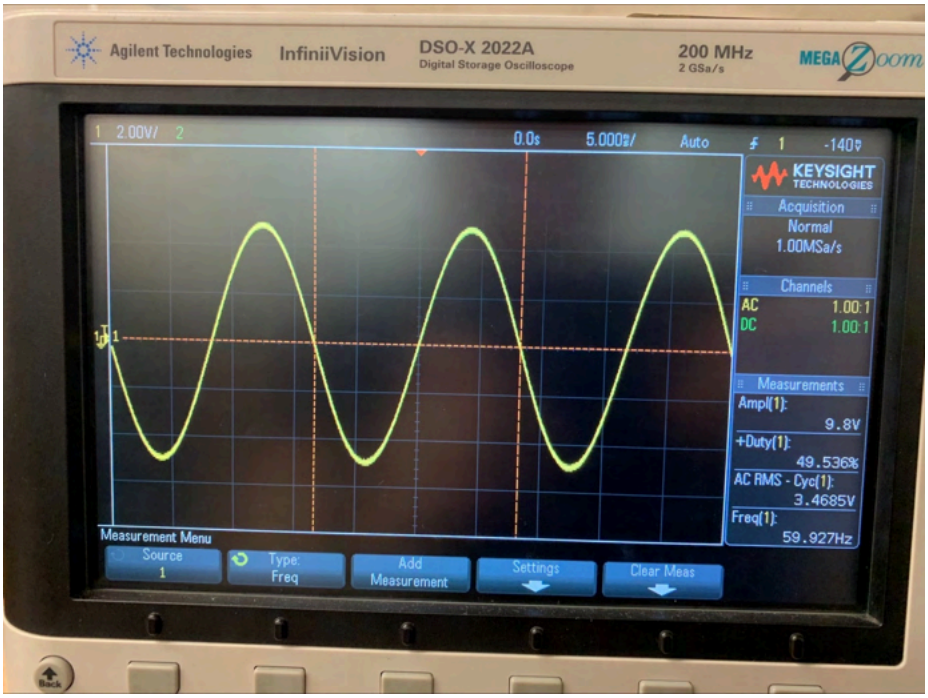

Figure 10B - Measuring the AC RMS value

In Figure 10A, we pressed the Type softkey in the Measurement menu and then selected AC RMS - N Cycles. In Figure 10B, we pressed the Add Measurement softkey and our AC-RMS value appeared on the right under the Measurements section. The AC-RMS value was 3.47 volts.

Conclusion

In the lab, we learned how to generate a sine wave function using a function generator by entering a specific frequency, peak-to-peak voltage, and offset voltage. By connecting this function generator to an oscilloscope, firstly we were able to visualize our sine wave function on the screen. Secondly, we learned to adjust the graphical properties of the wave by changing the vertical, horizontal, and trigger level knobs on the scope. Furthermore, we were able to identify the parameters and numbers displayed on the display such as the channel, vertical divisions, horizontal divisions, trigger source, and so on. Lastly, we learned how to manually measure the sine wave by adjusting cursors on the screen and automatically measure other data points by adding measurements. These basic functions will allow us to analyze AC circuits and other electronic waves in order to analyze them.